

13.3 Ethical Issues

Can the use of AI systems for software development be approved of from an ethical point of view? This question is perhaps asked a little late because we've almost reached the end of this book. The fact is that AI tools have more downsides than the occasional wrong answer. That is why we want to deal with this in the following sections.

Restricting the Use of AI for Coding

We've already mentioned this in the preface and would like to point it out again here: this book deals with AI exclusively in the context of software development.

Of course, we know that AI tools are used to generate fake news, images, and videos, and that AI algorithms have the potential to put entire professional groups out of work. AI is finding its way into more and more areas of our lives. The resulting problems will fundamentally change our society. These problems are undisputed, but like any AI application outside of coding, they aren't the subject of this book.

The same applies to the dystopian question of whether AI models will soon evolve into a superintelligence and then take over the world. This is still the stuff of science fiction books and films. The only worrying thing is how many renowned AI researchers see this as a real danger (read <https://righttowarn.ai>).

13.3.1 The Origin of AI Knowledge

Each of us three authors had already written other books before this one. Time and again, we've asked ourselves the following question: Were our books also used to train the language models whose application we describe here? Have our posts in discussion forums been used, our code

stored on GitHub, the articles in our blogs? These questions are difficult to answer. Most AI providers are very tight-lipped when it comes to a precise description of the training material.

Another question, however, is very easy to answer: What did we get in return for using our knowledge—in whatever form—to train LLMs?

Definitely nothing!

Microsoft, which in the past has always demanded very strict compliance with all conceivable (and inconceivable) rules on copyright and licensing issues, has surprisingly liberal ideas when it comes to AI training. Mustafa Suleyman, CEO for AI issues at Microsoft, put it this way: “I think that with respect to content that’s already on the open web, the social contract of that content since the ’90s has been that it’s fair use. Anyone can copy it, recreate with it, reproduce with it. That has been ‘freeware,’ if you like, that’s been the understanding” (6/26/2024). Naturally, this statement was met with a lot of criticism. Ultimately, however, Mr. Suleyman has only expressed what other AI companies are probably thinking to themselves.

There is no question that fair compensation for training material consisting of billions of pages of text and terabytes of data is virtually impossible. Who should get how much? How can the contribution of a text to the overall language model be understood? Tracing every single author would involve a huge amount of work. Think of sites like Stack Overflow with contributions from millions of developers or articles on Wikipedia that many authors have contributed to. But if you follow Microsoft’s way of thinking, Stack Overflow is a great freeware knowledge base for AI training. At the same time, the business model of Stack Overflow is being shaken by AI offerings. Is that “fair use”?

The difficulties in realizing fair knowledge compensation don’t change the fact that the entire AI business model has a massive downside. On one hand, there are IT companies worth billions that earn money with AI technology. On the other hand, there is the whole of humanity, which has created this knowledge.

Admittedly, many AI offerings on the web are available free of charge, albeit with certain restrictions. Nevertheless, we find it irritating that our

code stored on GitHub is used for AI training, while at the same time we have to pay again to use the model (also) trained with our code in GitHub Copilot.

There is no doubt that AI research, the training of AI models, and so on cost a lot of money. AI companies need a way to earn money, or further development is difficult. So far, in our capitalist world view, everything is somehow in order. It remains to be seen whether part of the revenue shouldn't flow back to where the knowledge was created or made accessible (schools, universities, publishers, etc.).

Sooner or later, courts will have to decide which data may be used for AI training and under what conditions. A legal dispute has already broken out over the use of GitHub code for AI training. As of December 2024, it looked as if the judges would side with Microsoft (read www.developer-tech.com/news/judge-dismisses-majority-github-copilot-copyright-claims).

13.3.2 Jobless IT Experts

Microsoft advertises GitHub Copilot with “55% faster coding.” If this were true, a company could lay off 35% of its software developers and give the rest a Copilot subscription. The downsized team would produce the same amount of code as before.

This calculation is misleading for various reasons, not least because software developers aren't just responsible for coding. They take on many other tasks where AI support is less or not at all relevant. Last but not least, this includes (human) contact with team members and customers.

NVIDIA's boss Jensen Huang is unimpressed. He goes even further and believes that software developers will be completely superfluous in the future. Of course, this statement is primarily due to marketing. Nevertheless, all IT experts must ask themselves whether they are making themselves partially redundant by using AI technologies. As authors, we should ask ourselves: Are we supporting this process with this book?

The Dead Live Longer

The risk of IT hordes seeking new jobs is definitely not dramatic, at least not as a result of the new AI tools. They aren't yet good enough for that. And even for the near future, we're skeptical that AI tools will replace a lot of real IT jobs. The following blog article sums up this view convincingly:

<https://dava.ai/n/hank/ai-didnt-take-my-job>.

However, we expect AI tools to massively change work in many IT areas. Well-trained, experienced developers will definitely have an advantage on the job market; ideally, they should be able to use AI tools efficiently.

What is interesting in this context is that managers have different, often completely exaggerated expectations than developers who actually work with AI tools. According to a study by Atlassian (www.atlassian.com/software/compass/resources/state-of-developer-2024), only just under 40% of respondents believe that current AI tools significantly improve their productivity. However, 60% still expect AI tools to significantly improve their productivity within the next two years.

In 2024, the annual Stack Overflow survey (<https://survey.stackoverflow.co/2024/ai#efficacy-and-ethics>) mentioned earlier in this chapter found out for the first time whether IT experts fear for their jobs because of AI tools: 68% don't see this risk, while 12% do (the rest are unsure).

13.3.3 Unlimited Energy Consumption

Cloud-based AI services start answering in a fraction of a second and present their answers faster than you can read along. You don't realize how much energy this requires until you try to run language models locally. The MacBook of one of the authors of this book was practically permanently idle during work. This only changed when Ollama had to execute requests to a language model locally: the power consumption immediately increased from 3 watts to 20 to 30 watts (in each case without screen, the MacBook was connected to an external monitor). However, these are still modest values compared to the power consumption of a "classic" PC with an NVIDIA graphics card: Although the computing power is then also suffi-

cient for significantly faster processing of larger voice models, the power consumption increases to several hundred watts. That's crazy!

The big AI companies will only smile wearily at the figures just presented. The data centers for running existing AI offerings and training new voice and multimedia models require as much power as entire small towns. Thanks to new hardware and improved optimization, it's hoped that the power consumption per prompt or for training new models will fall in the coming years. This is offset by increasing quality requirements and user numbers. Overall, it's to be feared that the demand for electricity for AI applications will continue to rise in the coming years.

Of course, every other IT service also needs electricity: the local operation of notebooks, monitors and Wi-Fi routers; storing data in the cloud; streaming and playing a video; and so on. But the power consumption of AI services goes beyond all previously known proportions. In view of the climate crisis, it's fair to ask to what extent AI services are sustainable or can become so in the future.